Recognition Kernel Completion of the Five-Label Weil Endpoint

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Abstract

This draft gives the reviewer-facing version of the Recognition Kernel Framework route to the five-label completed-Weil endpoint. The previous proof-lab presentation contained several overlapping paths; this version reorganizes them into one dependency chain and states the secondary certificates in appendices. The native chain is

positive-eta amplitude path equality → zero-as-cut endpoint discipline → common five-matrix seam integer → M_3

The endpoint inequality is the zero-cut projection-defect statement

$$K_0 = R_\Sigma^*(I - P_{\mathrm{src}})R_\Sigma \geq 0.$$

The paper separates this native endpoint from the classical membrane: the completed explicit formula and the Weil criterion are consumed only in the declared normalization. The framework is used as a typed anti-blindness ledger. Source restriction, blind kernels, normalization path ambiguity, representation dependence and zero-cut support loss are not metaphors; each is represented by a theorem node. The numerical and proof-lab evidence is summarized in certificate tables with repository pointers and Zenodo records.

Contents

1 Introduction

The purpose of this paper is to present the cleaned endpoint route after the MP proof-lab results were transferred into the Recognition Kernel Framework repository. The central correction is architectural. The proof no longer proceeds through a separate even-source positivity assertion. Instead, the endpoint is reached through the fixed five-label recognition carrier, the seam integer, the certified accepted block, the single-sector reduction and the zero-cut T03 completion.

The paper is written with a first-time reviewer in mind. The main theorem nodes are stated in the body. Proof-lab machinery that is useful but not terminal appears in appendices. Repository capsules are used as provenance, not as a substitute for spelling out the logical chain. The proof is still native in the framework sense: the classical explicit formula and the Weil criterion appear only at the final membrane.

1.1 The five main theorem nodes

The endpoint chain consumes the following nodes.

(T1) **Positive-eta amplitude path equality.** Normalization and eta continuation commute on the fixed five-label carrier for every positive eta.

(T2) **Zero-as-cut discipline.** Eta zero is treated as a support/cut event, not as an ordinary substitution into a positive-eta formula.

(T3) **Common five-matrix seam reduction.** The sign-sensitive obstruction is represented by one rank-at-most-five matrix whose threshold count is the seam integer.

(T4) **Accepted block positivity.** The transferred certificate gives $G_3 > 0$ and $M_3 > 0$, hence positivity of the accepted odd block.

(T5) **Zero-cut T03 endpoint.** The invariant source-Gram identity gives the projection defect $K_0 \geq 0$.

The appendix records the independent T18–T20 odd boundary-burden certificate. That certificate is important, but the body uses the T03 source-Gram route as the main endpoint closure.

2 Classical membrane and normalization contract

The paper separates the native construction from the classical terminal membrane.

Definition 2.1 (Test core and Fourier convention)**.** *Let $\mathcal{D} \subset C_c^\infty(\mathbb{R})$ be the logarithmic test core used for the completed-Weil form. For $f \in \mathcal{D}$, write*

$$\widehat{f}(z) = \int_{\mathbb{R}} f(x)e^{izx}\, dx,$$

with the understanding that the opposite Fourier sign conjugates the symmetric zero pairing and does not change the diagonal Hermitian sign once the convention is fixed throughout.

Definition 2.2 (Classical Weil membrane)**.** *For $f, h \in \mathcal{D}$, define the classical zero-side Hermitian form*

$$Q_W(f,h) = \sum_{\rho}^{\mathrm{sym}} \widehat{f}(z_\rho)\overline{\widehat{h}(\overline{z_\rho})}, \qquad z_\rho = \frac{\rho - \frac{1}{2}}{i},$$

where the summation is the symmetric completed-zero summation of the classical explicit formula.

Assumption 2.3 (Classical membrane in the declared normalization)**.** *The native boundary–Gamma–prime completed-Weil form W_ξ constructed on the recognition carrier is identified by the completed explicit formula with Q_W in the convention above. The classical Weil criterion is consumed in the same normalization:*

$$Q_W(f,f) \geq 0 \quad (f \in \mathcal{D}) \quad \iff \quad \text{RH}.$$

Remark 2.4. *The native part of the paper constructs and closes the endpoint sign of the completed-Weil form. The final implication to the Riemann hypothesis is exactly the classical membrane above. The paper does not claim that a primitive diagonal Euler determinant, the prime torus alone, or raw UGD holonomy independently replaces this membrane.*

3 Why the framework is not over-engineering

The Recognition Kernel Framework is introduced because ordinary operator notation can lose target-relevant information at the exact places where the endpoint proof must be stable. Table 1 records the obstruction ledger.

Failure mode	Framework repair
Source restriction	source-restriction rank formula and blind-kernel formula
Target-relevant blind kernel	minimal scalar lift or target-relative decoder
Path or normalization ambiguity	positive-eta amplitude path equality
Representation dependence	common five-matrix covariance
Accepted-block uncertainty	certified $G_3 > 0$ and $M_3 > 0$
Endpoint support loss	zero-cut T03 completion

Table 1: Each framework layer repairs a named obstruction.

Theorem 3.1 (Source restriction and minimal lift). *Let $S(q)P$ be a rich split representation and let B be a consumed source restriction. Put*

$$A(q) = BS(q)P, \qquad S(q) = \sum_s q^s D_s.$$

Then the all-parameter observable rank is

$$\operatorname{rank} \begin{pmatrix} BD_1P \\ BD_2P \\ \vdots \end{pmatrix},$$

and the blind kernel is

$$\bigcap_s \operatorname{Ker}(BD_sP).$$

If scalar supplemental observations G are added, the augmented map is injective exactly when

$$\operatorname{Ker} A \cap \operatorname{Ker} G = \{0\}.$$

Consequently at least $\dim \operatorname{Ker} A$ *scalar observations are necessary to repair a full blind defect. Target-relative repair requires only that the remaining blind kernel lie in the target kernel.*

Proof. The rank formula is the rank of the direct sum of all coefficient-level source restrictions. A vector is blind for the entire rich representation exactly when it is annihilated by every restricted coefficient map BD_sP, giving the displayed intersection. Adding a scalar observation map G removes the blind defect precisely when no nonzero vector is invisible to both the original restricted map and the supplement. This gives the injectivity criterion and the lower bound on the number of independent scalar repairs. □

Remark 3.2 (Endpoint meaning). *In the endpoint application, the source restriction is the passage to the target-relevant source readout, the blind kernel is the endpoint information not seen by that readout, and the five-label lift supplies the minimal repair. The theorem is therefore not decoration; it is the algebraic reason that a blind quotient and lift must be declared before endpoint promotion.*

4 Positive-eta amplitude analyticity

The positive-eta analytic theorem is one of the load-bearing results. It removes path-order ambiguity before the sign problem begins.

Definition 4.1 (Polar frame transport). *For a full-column-rank amplitude $\Theta = QH$, where $Q^*Q = I_5$ and $H > 0$, define the canonical transport between two polar frames Q_a, Q_b by*

$$U_{a\to b} = \operatorname{polar}(Q_b^* Q_a).$$

Theorem 4.2 (Fixed-five positive-eta path equality). *Let*

$$A_{\text{chart}} = AU, \qquad \Theta_\eta = Q_{\text{chart}} G_\eta^{1/2}, \qquad G_\eta = 1 \oplus (A_+ + \eta I_4), \quad \eta > 0.$$

Here A is the eta-independent raw amplitude, U is the fixed unitary accepted chart, Q_{chart} is the polar orientation of A_{chart}, and $G_\eta^{1/2}$ is strictly positive. Then

$$\operatorname{polar}(\Theta_\eta) = Q_{\text{chart}}$$

for every positive eta. Hence, for any positive eta values η_1, η_2,

raw eta transport $= I_5$, *normalized eta transport* $= I_5$, *raw-to-normalized transport* $= U^*$.

All positive-eta normalization/continuation rectangles are identity loops on the fixed five-label carrier.

Proof. Right multiplication by a unitary is polar-covariant, so $\mathrm{polar}(AU) = \mathrm{polar}(A)U$. Right multiplication by a strictly positive factor changes the metric scale but not the polar orientation. Since $G_\eta^{1/2} > 0$, the polar orientation of $Q_{\mathrm{chart}} G_\eta^{1/2}$ is Q_{chart}. Therefore the polar frame at every normalized positive-eta node is the same frame. The raw eta transport and normalized eta transport are identity maps on the label fibre, while the raw-to-normalized transport is the fixed chart unitary U^*. The two possible paths around any positive-eta normalization rectangle have the same product. □

Remark 4.3 (Finite rectangle shadow)**.** *The state-locked rectangle at* $\eta = 0.010$ *and* $\eta + h = 0.011$ *reported path transport difference* $3.0669860554696173 \cdot 10^{-15}$*, loop identity residual* $3.1396725820594018 \cdot 10^{-15}$*, defect-to-accepted leakage* $1.7140530928543942 \cdot 10^{-17}$*, endpoint scale difference zero, and no false checks. These numbers are the floating-point shadow of the exact factorization above.*

5 Zero as cut

The previous theorem says there is no positive-eta normalization holonomy. The endpoint issue is therefore the support event at eta zero. This is the role of the zero-as-cut theorem.

Definition 5.1 (Cut and joint cycles)**.** *The pre-cut carrier has one geometric cycle* γ_π*. The post-cut carrier has an independent cut-memory cycle:*

$$H_1(S^1; \mathbb{Z}) = \mathbb{Z}\gamma_\pi, \qquad H_1(T^2; \mathbb{Z}) = \mathbb{Z}\gamma_\pi \oplus \mathbb{Z}\mu_{\mathrm{cut}}.$$

On the ordered basis $(\gamma_\pi, \mu_{\mathrm{cut}})$*, set*

$$P_{\mathrm{joint}} = \mathrm{diag}(1, 0), \qquad P_{\mathrm{cut}} = \mathrm{diag}(0, 1).$$

Theorem 5.2 (Zero-as-cut projector theorem)**.** *The cut and joint projectors satisfy*

$$P_{\mathrm{cut}} P_{\mathrm{joint}} = P_{\mathrm{joint}} P_{\mathrm{cut}} = 0, \qquad P_{\mathrm{cut}} + P_{\mathrm{joint}} = I.$$

Hence no nonzero vector can be both a pure cut state and a complete joint state.

Proof. The identities follow immediately from the displayed diagonal matrices. If $u = P_{\mathrm{cut}} u = P_{\mathrm{joint}} u$, then

$$u = P_{\mathrm{cut}} P_{\mathrm{joint}} u = 0.$$

Thus simultaneous pure cut recognition and complete joint recognition force vanishing. □

Remark 5.3 (Visible return is not full return)**.** *For the join projection* $J = (1, 0)$ *and section* $S = (1, 0)^T$*, one has* $JS = I_1$*, but*

$$SJ = P_{\mathrm{joint}} \neq I_2.$$

Thus the visible circle coordinate can return while the full post-cut torus state does not. The exact memory defect is

$$I_2 - SJ = P_{\mathrm{cut}}.$$

This is the reason the endpoint is handled by a zero-cut source-Gram theorem rather than by an ordinary positive-eta continuation.

Remark 5.4 (Finite critical witness)**.** *The finite critical response packet recorded a surviving cut-memory coordinate with joint and cut-memory component norms both* $4.137400474477719 \cdot 10^{-5}$*, relative cut memory* 0.7071067811865472*, and complete state return false. This is a finite witness for the warning above: corrected target visibility does not automatically give full state return.*

6 Seam integer and the common five-matrix

The sign-sensitive obstruction is represented by one rank-at-most-five matrix. This section turns the seam language into a standard threshold-count statement.

Definition 6.1 (Seam charge)**.** *Let K_η be the compact seam response at positive shift eta. Define*

$$k_\Sigma(\eta) = \operatorname{rank} \mathbf{1}_{(1,\infty)}(K_\eta).$$

On the five-label source packet this is represented by

$$B_\eta = V_\eta^* S_{\eta,-}^{-1} V_\eta, \qquad \operatorname{rank} V_\eta \le 5,$$

and

$$k_\Sigma(\eta) = N_+(B_\eta - I).$$

Theorem 6.2 (Common five-matrix covariance)**.** *For a positive source operator S and a five-column source map V, set*

$$B = V^* S^{-1} V \in M_5(\mathbb{C}).$$

If $S^U = USU^*$ and $V^U = UV$, then

$$(V^U)^*(S^U)^{-1}V^U = B.$$

A unitary change of the five labels conjugates B. Therefore the spectrum, threshold projection, seam charge and sign decision are representation invariant.

Proof. Since $(USU^*)^{-1} = US^{-1}U^*$,

$$(V^U)^*(S^U)^{-1}V^U = V^*U^*US^{-1}U^*UV = V^*S^{-1}V.$$

If the label basis is changed by a unitary U_5, then V is replaced by VU_5, and the matrix becomes $U_5^* B U_5$. Hence threshold ranks and spectral projections are invariant. □

Corollary 6.3 (Matrix, seam and threshold holonomy)**.** *The native matrix, prime-torus matrix and threshold-holonomy presentation are faithful avatars of the same obstruction:*

$$B_\eta \le I \iff k_\Sigma(\eta) = 0 \iff H_{\text{threshold}}(\eta) = I,$$

where

$$H_{\text{threshold}}(\eta) = I - 2\mathbf{1}_{(1,\infty)}(B_\eta).$$

Thus the prime torus is a chart of the same matrix, not a second independent proof route.

7 The accepted block and M_3

The finite accepted odd block is

$$E_3 = \operatorname{span}\{u, |x|u, |x|^2u, e_3\}.$$

Let G_3 be the completed-Weil Gram on $V_3 = \operatorname{span}\{u, |x|u, |x|^2u\}$, and let b_3 be the coupling vector from V_3 to e_3. Define

$$M_3 = q(e_3, e_3) - b_3^* G_3^{-1} b_3.$$

Theorem 7.1 (Certified accepted block)**.** *The transferred M3 certificate gives*

$$\lambda_{\min}(G_3) = 2.0838177120437723 \cdot 10^{-13} > 0$$

and

$$3.08520376681449 \cdot 10^{-13} < M_3 < 2.993597611951253 \cdot 10^{-12}.$$

Therefore $M_3 > 0$. Since M_3 is the Schur complement of G_3 in the bordered Gram, the completed-Weil form is positive definite on E_3.

Proof. The block Gram on $E_3 = V_3 \oplus \mathbb{C}e_3$ has the bordered form

$$\begin{pmatrix} G_3 & b_3 \\ b_3^* & q(e_3, e_3) \end{pmatrix}.$$

For a Hermitian bordered matrix with $G_3 > 0$, positivity is equivalent to positivity of the Schur complement

$$q(e_3, e_3) - b_3^* G_3^{-1} b_3 = M_3.$$

The certificate gives both $G_3 > 0$ and $M_3 > 0$, so the accepted block is positive definite. The certificate is correlated: the self channel, coupling vector, Schur compensation and prime/Gamma truncation are enclosed together; independent coordinate boxes for b_3 are not used. □

8 Single-sector terminal reduction

Both parity sectors remain represented in the bilateral carrier. The single-sector statement is a later terminal reduction, not a deletion of the even sector.

Assumption 8.1 (Reduction inputs)**.** *The following transferred inputs are consumed.*

(i) The representation-complete bilateral carrier preserves the completed-Weil target.

(ii) The seam-integer/common five-matrix theorem identifies the sign-sensitive terminal obstruction with the five-label threshold packet.

(iii) The common source-sign theorem and favorable even-boundary reduction remove the even sector as an independent terminal gate.

(iv) The odd five-label packet is the remaining target-relevant endpoint obstruction.

Theorem 8.2 (One-sector terminal sufficiency)**.** *Under the reduction inputs, the completed endpoint sign is equivalent to the nonnegativity of the odd five-label endpoint packet. The even sector is retained upstream but is not an independent terminal gate after the reduction.*

Proof. By representation completeness, no target-relevant completed-Weil information is lost by passing to the bilateral carrier. The seam-integer theorem expresses the sign-sensitive obstruction as the threshold rank of the common five-matrix. The source-sign and favorable even-boundary reduction place the even contribution in the accepted/non-terminal part of the carrier. Therefore a nonnegative endpoint fails only if the odd five-label threshold packet has a positive threshold count. Conversely, if that packet is nonnegative, the common five-matrix has no super-threshold direction and the endpoint obstruction vanishes. Thus the odd five-label packet is terminal after, and only after, these reductions. □

9 Zero-cut T03 endpoint completion

The T03 theorem is the main endpoint route used in the paper. It replaces orientation-dependent ambient equality by an invariant source-Gram identity.

Definition 9.1 (Endpoint source-Gram data)**.** *Let R_Σ be the total five-label response readout, P_{src} the source projection, $S^{\mathrm{full}}_{0,-}$ the complete unshifted odd source, and W_Σ the five-label response map. The endpoint source-Gram target is*

$$R_\Sigma^* P_{\mathrm{src}} R_\Sigma = \big((S^{\mathrm{full}}_{0,-})^{\dagger/2} W_\Sigma\big)^* \big((S^{\mathrm{full}}_{0,-})^{\dagger/2} W_\Sigma\big).$$

Theorem 9.2 (Six-block convergence criterion)**.** *Suppose finite transported source maps give finite Gram defects Δ_n satisfying*

$$\|\Delta_n\| \le \varepsilon_{\mathrm{vis}}(n) + \varepsilon_{\mathrm{blind}}(n) + \varepsilon_{\mathrm{cross}}(n) + \varepsilon_{\mathrm{tail}}(n) + \varepsilon_{\mathrm{quad}}(n) + \varepsilon_{\mathrm{chart}}(n),$$

where the six terms respectively bound visible compression error, five-dimensional blind response error, both cross blocks, omitted prime/Gamma tails, quadrature/window/solve/rounding errors, and native/Feshbach/prime-torus chart transport. If

$$\varepsilon_{\mathrm{vis}}(n) + \varepsilon_{\mathrm{blind}}(n) + \varepsilon_{\mathrm{cross}}(n) + \varepsilon_{\mathrm{tail}}(n) + \varepsilon_{\mathrm{quad}}(n) + \varepsilon_{\mathrm{chart}}(n) \longrightarrow 0,$$

then the invariant source-Gram identity holds.

Proof. Each finite transported source map produces a finite Gram comparison between the total five-label response readout and the source-projected response. The six listed errors form an outward norm bound on the full Gram defect. Since the total bound tends to zero, $\Delta_n \to 0$ in operator norm on the finite label space. Passing to the limit gives equality of the two limiting Gram matrices. Equal Gram matrices determine a canonical partial isometry between final ranges, so literal equality of ambient source maps is unnecessary. □

Theorem 9.3 (Zero-cut projection-defect positivity)**.** *Assume the invariant source-Gram identity and the independently constructed total-response Gram*

$$R_\Sigma^* R_\Sigma = \mathrm{diag}(1, A_3).$$

Then

$$K_0 = R_\Sigma^*(I - P_{\mathrm{src}})R_\Sigma \ge 0.$$

The endpoint Schur equivalence transfers this projection-defect positivity to the completed-Weil endpoint sign in the declared normalization.

Proof. Since P_{src} is an orthogonal projection, $I - P_{\mathrm{src}} \ge 0$. Hence for any label vector $c \in \mathbb{C}^5$,

$$\langle c, K_0 c\rangle = \langle R_\Sigma c, (I - P_{\mathrm{src}})R_\Sigma c\rangle \ge 0.$$

The invariant source-Gram identity identifies the source-visible part of the endpoint response with the complete unshifted source readout. The remaining projection defect is exactly the zero-cut endpoint obstruction. The Schur endpoint equivalence therefore yields the completed-Weil endpoint sign. □

10 Main theorem

Theorem 10.1 (Completed-Weil endpoint theorem)**.** *Assume the native completed-Weil carrier is in the declared normalization and consume Theorems 4.2, 5.2, 6.2, 7.1, 8.2 and 9.3. Then the completed-Weil endpoint sign holds on the target-faithful completed carrier.*

Proof. The positive-eta path theorem removes normalization ambiguity on the fixed five-label carrier. The zero-as-cut theorem identifies eta zero as a support/cut event rather than an ordinary path-holonomy issue. The common five-matrix theorem reduces the sign-sensitive obstruction to the representation-invariant threshold matrix B_η and seam charge k_Σ. The M_3 theorem proves positivity of the accepted odd block. The single-sector theorem identifies the remaining terminal packet with the odd five-label source obstruction after the seam/source/even reduction. Finally, the T03 theorem supplies the invariant source-Gram identity and hence

$$K_0 = R_\Sigma^*(I - P_{\mathrm{src}})R_\Sigma \geq 0.$$

The endpoint Schur equivalence transfers this projection-defect positivity to the completed-Weil endpoint sign. □

Corollary 10.2 (Classical terminal implication)**.** *Under the classical membrane in Section 2, the completed-Weil endpoint theorem implies the Riemann hypothesis.*

Proof. The native theorem gives nonnegativity of the completed-Weil form in the declared normalization. The completed explicit formula identifies this form with the symmetric zero-sum Weil form Q_W. The classical Weil criterion then gives RH. □

11 Numerical certificates and reproducibility

The proof contains certified numerical inputs. The main numerical evidence is summarized in Table 2. The reproducibility archive is recorded as DOI `10.5281/zenodo.21760119`; the unified publication record is recorded as DOI `10.5281/zenodo.21735284`. The repository theorem capsules are in `theorum/`, and reusable proof machinery is in `mp_gold/`.

Table 2: Certificate table for reviewer entry.

Node	Evidence	Claim used in paper
Positive-eta path equality	finite rectangle at 0.010 → 0.011 plus exact polar factorization	no positive-eta normalization holonomy
Zero-as-cut	cut/joint projector algebra; finite critical memory witness	eta zero is a cut/support event
Common five-matrix	covariance theorem and threshold holonomy equivalence	seam integer is a threshold count
M3 accepted block	$3.08520376681449 \cdot 10^{-13} < M_3 < 2.993597611951253 \cdot 10^{-12}$	accepted odd block positive
T03 endpoint	six-block Gram decomposition with vanishing total error	invariant source-Gram identity and $K_0 \geq 0$
T19–T20 appendix	$\beta_\partial^{\mathrm{cut}} \leq 0.9998319617060448 < 1$ and source injectivity	independent strict odd certificate

Remark 11.1 (Why the decimals are not the theorem)**.** *The numerical values are used only through directed certificates: intervals, outward error bounds, exact algebraic identities, and fail-closed tests. The paper does not use a fitted endpoint extrapolation or independent b_3 boxes to prove M_3.*

12 Resolution of structural objections

The revised paper answers the main reviewer objections as follows.

- Normalization and eta-continuation commute by positive-eta amplitude factorization.
- Eta zero is not treated as a naive substitution; it is a cut/support event.
- The seam integer is the threshold count of one common five-matrix.
- The prime torus is a faithful chart of that same matrix.
- M_3 is certified positive by a correlated Schur interval.
- One sector is terminal only after the seam/source/even reduction; the even sector is not erased.
- The endpoint source map is handled by an invariant Gram identity rather than orientation-dependent ambient equality.
- The framework is not over-engineering because each layer repairs a named source, blind-kernel, path, representation or endpoint defect.

13 Claim boundary

The paper claims the transferred RKF-native endpoint chain and clearly marks the classical membrane. It does not claim that raw UGD, the prime torus, or a diagonal Euler determinant independently proves the Riemann hypothesis. It does not claim that the even sector is absent. It does not promote numerical evidence unless the corresponding directed certificate is part of the theorem chain.

14 Conclusion

The cleaned architecture is deliberately short:

positive-eta path equality $\to$ zero-as-cut $\to$ five-matrix seam integer $\to$ $M_3 > 0$ $\to$ single-sector endpoint $\to$ K_0

The framework is needed because the endpoint is a target-faithful completion problem. Once source restriction, blind kernels, normalization ambiguity, sector terminality and zero-cut support are typed, the terminal object is the zero-cut projection defect. The classical explicit formula and Weil criterion then form the final membrane.

A Independent odd boundary-burden certificate: T18–T20

This appendix records a strong supporting theorem that is not the main T03 route. It gives an independent odd-sector burden certificate.

A.1 Boundary-burden equivalence

T18 identifies the odd completed-Weil sign with the cut boundary burden:

$$\text{full odd sign} \quad \Longleftrightarrow \quad \beta_\partial^{\mathrm{cut}} \le 1.$$

Equivalently,

$$S_{0,-}^{\mathrm{full}} - L_\partial^\dagger L_\partial \ge 0$$

once the boundary burden is at most one.

A.2 T19 refined endpoint inequality

For positive cut events with source energies s_j and boundary masses m_j, T19 uses

$$\beta_0 - \beta_\eta = \sum_j m_j \frac{\eta}{s_j(s_j+\eta)}.$$

The reconstructed Arb profile uses 2400 active cells at step 0.005, with 78734 prime-power events. At

$$\eta = 1.5625 \cdot 10^{-5}, \qquad \beta_\eta^+ = 0.999773667408463,$$

the correction ledger gives

$$\beta_\partial^{\mathrm{cut}} \le 0.9998319617060448 < 1.$$

Thus the full odd completed-Weil sign is closed on the declared native source carrier.

A.3 T20 strict odd positivity

T20 adds source injectivity:

$$\operatorname{Ker} S_{0,-}^{\mathrm{full}} = \{0\}.$$

Combining this with the strict reserve

$$1 - \beta_\partial^{\mathrm{cut}} \ge 0.0001680382939552$$

gives strict positivity on every nonzero odd core state:

$$\langle F, W_3^- F\rangle > 0, \qquad F \ne 0.$$

A.4 Relation to the main route

The T03 route and the T19–T20 route are not competing endings:

T03 route	invariant source-Gram identity $\Rightarrow K_0 \ge 0$,
T19–T20 route	$\beta_\partial^{\mathrm{cut}} < 1 \Rightarrow$ strict odd sign.

The body keeps T03 as the terminal spine and records T19–T20 as an independent odd-sector certificate.

B Auxiliary certified theorem nodes

B.1 Channel-Hermite no-escape

The X_3 channel-Hermite Parseval frame uses the analysis map

$$C_3 f = \left(e^{3|x|/2} f,\ (2\pi)^{-1/2}\sqrt{1+|G|}\,\widehat{f}\right)$$

with

$$\|C_3 f\|^2 = \|f\|_{X_3}^2.$$

It proves parity-resolved global no-escape and projected-frame completeness after accepted-memory removal. This supports response completeness but is not the terminal endpoint theorem.

B.2 Channel compatibility projection

The native analysis range is the closed graph of

$$T = B_0 A_0^{-1}.$$

Raw split-channel packets must be projected to the compatible graph before they update native couplings. The warning certificate is

$$\text{raw channel pairing} = 15/4 > 0, \qquad \text{native compatible pairing} = -1/4 < 0.$$

Thus raw channel sign cannot be promoted without compatibility projection.

B.3 Curvature-to-seam index law

The lawful curvature relation is spectral-flow typed:

$$k_\Sigma(\eta; u_1) - k_\Sigma(\eta; u_0) = -(\mathsf{I} - \mathsf{K}_\eta(\mathsf{u_s}); 0).$$

A scalar thermodynamic ratio is not canonically equal to the integer seam charge. It can influence the seam only through an explicit compact-operator path.

B.4 Unified seam block-Krylov machinery

For a finite-rank seed map S, define

$$V_m = \operatorname{span}\{K_{\Sigma,\eta}^j \operatorname{Ran} S : 0 \le j \le m\}.$$

The same-projection two-block upper bound gives

$$\sup \sigma(K) \le \frac{a + d + \sqrt{(a-d)^2 + 4b^2}}{2}$$

when a, b, d are computed against the same projection. This is future compact-operator machinery, not a substitute for the T03 endpoint.

C Repository provenance

The main theorem capsules live in `theorum/`. Supporting results are summarized in `mp_gold/`. The source PRs are recorded in `theorum/provenance.json` and `mp_gold/gold_manifest.json`. Raw verifier scripts, state capsules and large numerical ledgers remain outside this public paper draft unless explicitly archived.